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Inventor(s)	McCoy; Craig et al.

Fluid vaporizers

Abstract

A vaporizer apparatus includes a liquid inlet; a plurality of plates in a stacked arrangement and configured with openings and surfaces defining a tortuous path for a liquid through the plurality of plates; an atomizer configured to atomize liquid from the liquid inlet and inject a spray of the liquid toward a first plate of the plurality of plates, wherein the plurality of plates is configured to direct a flow of the liquid through the tortuous path from a first end corresponding to the first plate and output a vapor of the liquid at a second end of the tortuous path corresponding to a last plate of the plurality of plates; and at least one heating element configured to heat the plurality of plates to vaporize the liquid along its flow through the tortuous path and generate the vapor.

Inventors: McCoy; Craig (San Jose, CA), Laaksonen; Tapani (Fremont, CA), Kudriavtsev; Vladimir (Palo Alto, CA), Garvey; Jose (San Jose, CA)

Applicant: Yield Engineering Systems Inc. (Fremont, CA)

Family ID: 1000008337237

Assignee: Yield Engineering Systems, Inc. (Fremont, CA)

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Primary Examiner: Chou; Jimmy

Attorney, Agent or Firm: Fish & Richardson P.C.

Background/Summary

TECHNICAL FIELD

(1) The present disclosure generally relates to material processing systems, e.g., fluid vaporizers for vaporizing liquids.

BACKGROUND

(2) Vapor can be delivered to process chambers for various processes such as etching and material deposition. The vapor can be obtained by vaporizing a liquid.

SUMMARY

(3) Implementations of the present disclosure are generally directed to systems, methods, and devices associated with fluid and materials processing, e.g., fluid vaporizers for vaporizing liquids.

(4) Some aspects of this disclosure relate to a vaporizer apparatus that includes: a liquid inlet; a plurality of plates in a stacked arrangement and configured with openings and surfaces defining a tortuous path for a liquid through the plurality of plates; and an atomizer configured to atomize liquid from the liquid inlet and inject a spray of the liquid toward a first plate of the plurality of plates. The plurality of plates is configured to direct a flow of the liquid through the tortuous path from a first end corresponding to the first plate and output a vapor of the liquid at a second end of the tortuous path corresponding to a last plate of the plurality of plates. The vaporizer apparatus includes at least one heating element configured to heat the plurality of plates to vaporize the liquid along its flow through the tortuous path and generate the vapor.

(5) This and other vaporizer apparatuses described herein can have one or more of at least the following characteristics.

- (6) In some implementations, the plurality of plates are stacked with intervening spaces having a specified dimension.
- (7) In some implementations, the plurality of plates include two plates in succession along the tortuous path. The two plates are oriented parallel to one another and spaced apart along a stack direction. The two plates include respective openings arranged at different lateral positions.
- (8) In some implementations, the respective openings of a first plate of the two plates are arranged more radially inward compared to the respective openings of a second plate of the two plates.
- (9) In some implementations, the first plate has a surface exposed to the spray of the liquid.
- (10) In some implementations, the plurality of plates are arranged in a first stack and a second stack spaced laterally apart from the first stack. The first plate is closest to the liquid inlet among plates of the first stack. The last plate is closest to the liquid inlet among plates of the second stack.
- (11) In some implementations, the plurality of plates includes a first plurality of plates stacked in a first section, and a second plurality of plates stacked in a second section. The first plurality of plates is laterally spaced apart from the second plurality of plates. The first plurality of plates includes openings and surfaces defining a first portion of the tortuous path. The second plurality of plates includes openings and surfaces defining a portion section of the tortuous path.
- (12) In some implementations, the vaporizer apparatus includes a divider separating the first plurality of plates from the second plurality of plates. The divider includes an opening that defines a portion of the tortuous path.
- (13) In some implementations, the vaporizer apparatus includes a divider separating the first plurality of plates from the second plurality of plates. The divider includes a slot to receive a first heating element of the at least one heating element.
- (14) In some implementations, (i) the first plurality of plates and the second plurality of plates include different numbers of plates, (ii) a total dimension of the first plurality of plates along a stack direction is different from a total dimension of the second plurality of plates along the stack direction, or (i) and (ii).
- (15) In some implementations, the plurality of plates includes a first type of plate and a second type of plate. The first type of plate and the second type of plate have openings in different lateral positions, and the first type of plate and the second type of plate alternate along the tortuous path.
- (16) In some implementations, the vaporizer apparatus includes an enclosure in which the plurality of plates are arranged. An inner sidewall of the enclosure defines the tortuous path in conjunction with the plurality of plates.
- (17) In some implementations, the at least one heating element includes at least one cartridge heating extending through two or more of the plurality of plates.
- (18) In some implementations, the tortuous path includes: a first portion extending from the first end of the tortuous path with a first overall flow direction; and a second portion extending from the first portion to the second end of the tortuous path with a second overall flow direction. The second overall flow direction is opposite to the first overall flow direction.
- (19) In some implementations, the tortuous path switches from a first direction to a second direction opposite the first direction at least twenty times.
- (20) In some implementations, the tortuous path is defined in a volume having an overall dimension D along a direction, and the tortuous path has a length at least $3D$ along the direction.
- (21) In some implementations, the vaporizer apparatus is configured to deliver the vapor to a chamber in the absence of a carrier gas.
- (22) In some implementations, gas output from the vaporizer apparatus is at least 95% vapor of the liquid.
- (23) In some implementations, the liquid inlet is the only fluid inlet of the vaporizer apparatus.
- (24) In some implementations, the plurality of plates are separate components from one another.
- (25) Some aspects of this disclosure relate to a material processing system that includes: a vessel configured to store a liquid; a pressurizer configured to pressurize the liquid in the vessel; and a

vaporizer fluidically coupled to the vessel. The vaporizer includes: a liquid inlet configured to receive the pressurized liquid from the vessel, and a plurality of plates in a stacked arrangement and configured with openings and surfaces defining a tortuous path for the liquid through the plurality of plates. The plurality of plates is configured to direct a flow of the liquid from the liquid inlet through the tortuous path from a first end corresponding to a first plate of the plurality of plates and output a vapor of the liquid at a second end of the tortuous path corresponding to a last plate of the plurality of plates. The vaporizer includes at least one heating element configured to heat the plurality of plates to vaporize the liquid along its flow through the tortuous path and generate the vapor. The material processing system includes a chamber fluidically coupled to the second end of the tortuous path and configured to receive the vapor to perform a material processing operation in the chamber.

(26) This and other material processing systems described herein can have one or more of at least the following characteristics.

(27) In some implementations, the material processing system includes: a fluid valve connected between the vaporizer and the vessel; and a controller configured to pulse the fluid valve to provide the liquid into the vaporizer. The controller is configured to pulse the fluid valve with an on-time in a range from 20 ms to 250 ms and an off-time in a range from 0.5 seconds to 4.0 seconds.

(28) In some implementations, the pressurizer is configured to pressurize the liquid with a pressure in a range from 35 psi to 90 psi.

(29) In some implementations, the vessel is configured to store formic acid as the liquid.

(30) In some implementations, the vaporizer includes an atomizer configured to atomize liquid from the liquid inlet and inject a spray of the liquid toward the first plate.

(31) In some implementations, the plurality of plates includes two plates in succession along the tortuous path. The two plates are oriented parallel to one another and spaced apart. The two plates include respective openings arranged at different lateral positions.

(32) Some aspects of this disclosure relate to a vaporizer apparatus that includes: a plurality of plates in a stacked arrangement and configured with openings and surfaces defining a tortuous path for a liquid through the plurality of plates. The plurality of plates is configured to direct a flow of the liquid through the tortuous path from a first end corresponding to a first plate of the plurality of plates and output a vapor of the liquid at a second end of the tortuous path corresponding to a last plate of the plurality of plates. The vaporizer apparatus includes at least one heating element configured to heat the plurality of plates to vaporize the liquid along its flow through the tortuous path and generate the vapor.

(33) The details of one or more implementations of the subject matter of this disclosure are set forth in the accompanying drawings and the description below. Other features, aspects, and advantages of the subject matter will become apparent from the description, the drawings, and the claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a diagram illustrating an example of a material processing systems.

(2) FIG. 2 is an exploded view of an example of a vaporizer.

(3) FIG. 3A is a perspective view of an example of a baffle assembly.

(4) FIG. 3B is a cross-sectional view of an example of a vaporizer.

(5) FIG. 3C is a diagrammatic top view of an example of a vaporizer.

(6) FIG. 3D is a sectional view of an example of a vaporizer.

(7) FIG. 4 is an exploded view of examples of insulating jackets.

(8) FIGS. 5A-5B are top views of examples of plates of a baffle assembly.

- (9) FIG. 6 is a perspective view of an example of a vaporizer box.
(10) FIG. 7 is a diagram illustrating an example of a pressurizer.
(11) FIG. 8 is a diagram illustrating an example of a liquid injection process.
(12) FIG. 9 is a diagram illustrating an example of material processing.

DETAILED DESCRIPTION

(13) Various aspects of this disclosure describe systems, methods, and devices for materials processing, e.g., fluid vaporizers for vaporizing liquids. An efficient vaporizer can include features (e.g., an atomizer and/or a tortuous path) that result in high flow rates of output vapor, for example, even in the absence of a carrier gas.

(14) As an illustrative example, some aspects of this disclosure can be applied to the use of a suitable chemical fluid (e.g., formic acid, HCOOH , among others) for solder reflow. In a solder reflow process, electrical components attached to a substrate (e.g., a circuit board) by a solder paste are placed in a chamber (e.g., an oven) and heated to “reflow” the solder (converting the solder into a molten state) and reform the solder joints, making the joints stronger and permanent. Flux-free reflow processing (providing improved bonds compared to processes that use a flux) can be performed at relatively low temperatures in a formic acid atmosphere. The formic acid vapor chemically reacts with metal oxides (e.g., surface oxides on a metal pad) to form an oxide-free bond. The use of formic acid can reduce or eliminate the need for pre-reflow fluxing and post-reflow flux removal, and provides effective oxide removal at low temperatures for an overall flexible process. For example, a formic acid atmosphere during solder reflow can effectively remove oxide layers from tin.

(15) The effectiveness of formic acid may be a function of a concentration (partial pressure) of formic acid in the chamber. For example, to ensure oxide removal and target processing characteristics, processing (e.g., heating) may be performed only when the chamber has reached a certain minimum partial pressure of formic acid, or has a partial pressure of formic acid within a defined range. However, as circuit system sizes (and, correspondingly, chamber sizes) increase, it may be challenging to achieve target formic acid vapor concentrations while maintaining desired throughput. For example, as elements of the electronics industry transition to the use of glass panels as substrates for heterogeneous integration, chambers with volumes of about 120 L may be used to accommodate systems undergoing solder reflow. For example, the panels may be up to 65 cm in length and width, necessitating large chambers. Conventional vapor delivery systems may provide about 1 mg/s of liquid, a rate incompatible with high-throughput processing using such large chambers.

(16) Moreover, conventional vapor delivery systems may be reliant on the use of a carrier gas, e.g., for liquid/vapor transport and/or for liquid atomization prior to vaporization. However, the presence of a carrier gas directly dilutes the vapor concentration and may be undesirable for chemical reaction processes. Further, the presence of the carrier gas may complicate concentration control, e.g., due to the confounding contribution of the carrier gas on pressure or mass-flow measurements. Also, many conventional vapor delivery systems rely on mass flow controllers (MFC) for mass flow control, resulting in large system volumes for both the MFC and other components (e.g., vaporizers) and complicating overall system design. Conventional vapor delivery systems tend to be low-power, use short paths, and follow limited volumetric vaporization rate designs.

(17) Some vaporizers and associated systems described herein can be used to obtain significantly higher delivery rates of chemical vapors, e.g., flow rates of at least 2 g/s. As such, these systems can be used, for example, for high-throughput solder reflow using large-area substrates in large chambers. In some implementations, the systems described herein can advantageously operate without reliance on or use of a carrier gas. In some implementations, mass/flow control can be provided using pulsed volume injection control.

(18) The systems described herein are also applicable to other manufacturing and fabrication

processes, such as annealing in an atmosphere, etching, chemical vapor deposition, and atomic layer deposition. For example, the vaporized liquid can be tetramethylammonium hydroxide (TMAH) for use in atomic layer deposition of aluminum oxide. As another example, the vaporized liquid can be titanium chloride (TiCl₄) for chemical vapor deposition of titanium nitride. As another example, the vaporized liquid can be tetrakis(ethylmethylamino)hafnium(IV) (TEMAH) for atomic layer deposition of hafnium oxide. As a further example, the systems described herein are well-suited for vaporization of silanes (e.g., Si—Cl, Si—OCH₃, Si—OCH₂CH₃, and the like), e.g., for use in deposition and coating processes. In some implementations, the vapor generated by the vaporizer **102** is an etchant for use in etch processes.

(19) FIG. **1** illustrates an example of a material processing system **100** according to some implementations of the present disclosure. The system **100** includes a vessel **104** for holding a liquid, a vaporizer **102** for vaporizing the liquid to obtain a vapor, and a chamber **112** into which the vapor is provided for one or more processes. The vaporizer **102** is described in more detail with respect to FIGS. **2** through **6**. The vessel **104**, pressurizer **106**, and valve **116** are described in more detail with respect to FIG. **7**. In some implementations, a pressurizer **106** is included to pressurize the liquid in the vessel **104**, to force the liquid into the vaporizer **102**. A valve **116** can be controlled (e.g., by a controller **124**) to control the flow of liquid from the vessel **104** into the vaporizer **102**.

(20) The controller **124** is configured to receive signals indicative of physical parameters of the material processing system **100** (e.g., pressure(s), temperature(s), and/or the like), and to send signals that control elements of the material processing system **100**. For example, as discussed below, the controller **124** can send signals to heaters to set temperature(s) of the material processing system **100**, to a regulator **704** (shown in FIG. **7**) to set a push pressure of the pressurizer **106**, and/or to the valve **116** to enable and disable liquid flow into the vaporizer **102**. The controller **124** can include one or more computing systems local to and/or remote from other components of the material processing system **100**.

(21) Further discussion of the material processing system **100** of FIG. **1**, and processes performed using the material processing system **100**, are provided with respect to FIG. **9**.

(22) FIG. **2** is an exploded view of an example of the vaporizer **102**. In this example, the vaporizer **102** includes a cover **202**, a baffle assembly **204**, and an enclosure (alternatively referred to as a body) **206**. The baffle assembly **204** fits into an aperture **228** in the enclosure **206** to be circumferentially surrounded by and enclosed in the enclosure **206**, and a cover plate **212** of the cover **202** fits onto a flange **226** of enclosure **206** to substantially seal the aperture **228**. Screws **220** extend through screw holes **216** in the cover plate **212** and into the flange **226** to hold the cover **202** and the enclosure **206** together. A retainer **214** (e.g., a vertical stop) extending downward from the cover plate **212** restricts movement of the baffle assembly **204**, e.g., prevents the baffle assembly **204** from sliding if the vaporizer **102** is turned upside-down after assembly. A thermocouple feedthrough **213** permits the insertion of a thermocouple to measure the internal temperature of the vaporizer **102**, e.g., while retaining a sealed environment for the vaporizer **102**.

(23) As described in greater detail with respect to FIGS. **3A-3B**, the baffle assembly **204** includes a plurality of baffles, fins, or blades, which subsequently are referred to, without loss of generality, as plates **304** such as plates **304-1** and **304-2**. These baffles, fins, blades, or plates can be configured in a stacked arrangement, positioned on top of one another with intervening spaces on the order of millimeters, or another spacing. Openings in the baffles, fins, blades, or plates define the tortuous path. For example, as discussed in further detail below, openings in adjacent baffles, fins, blades, or plates can be laterally offset with respect to one another, such that the tortuous path **108** includes repeated twists in order to pass through successive openings.

(24) A thermocouple slot **224** formed in an outer surface of the enclosure **206** is configured to receive a thermocouple for measuring a temperature of the vaporizer **102**. For example, the thermocouple slot **224** can receive an “over temperature thermocouple” which outputs temperature data used by a controller (e.g., controller **124**) to determine whether the vaporizer **102** is exceeding

a target or safe temperature range. One or more other temperature sensors of any suitable type (e.g., resistive and/or diode-based) can instead or additionally be included elsewhere in the vaporizer **102**. For example, in some implementations, a thermocouple is included in a central heater slot **222** to measure and output a temperature of the vaporizer **102**. In some implementations, an internal thermocouple of the vaporizer **102** is connected externally through the thermocouple feedthrough **213**. The temperature sensor(s) (e.g., thermocouple(s)) can provide signals indicative of sensed temperatures to the controller **124**, and the controller **124** can adjust the power of one or more heaters (such as cartridge heaters **218**, **234**, and/or **344**) to cause the temperature to be within a target range, as discussed below.

(25) The vaporizer **102** includes at least one heater for heating the vaporizer **102** (e.g., heating walls of a tortuous path in the vaporizer **102**). As discussed below, the operation of the at least one heater can vaporize liquid injected into the vaporizer **102**. In the example of FIG. 2, the vaporizer **102** includes six cartridge heaters extending longitudinally through an interior of the enclosure **206** and through corresponding heater slots in the baffle assembly **204**. Each hemisphere (or section, or chamber) of the baffle assembly **204** (see the discussion of FIGS. 3A-3B for discussion of the hemispheres or sections) includes three heater slots **232** through which a corresponding cartridge heater **218** extends (for example, through multiple plates), and a central cartridge heater **234** extends through the central heater slot **222** in a center of the baffle assembly **204**. The cartridge heaters **218**, **234** can be controlled to heat the baffle assembly **204** to a target temperature. For example, in some implementations, the at least one heater of the vaporizer **102** is controlled (e.g., by the controller **124**) to heat the vaporizer **102** to a temperature in a range from 90° C. to 220° C., e.g., in a range from 90° C. to 150° C., or in a range from 150° C. or 160° C. to 220° C., for example, in a range from 170° C. to 200° C. In some implementations, the temperature is controlled to be less than a maximum rating of o-ring seals of the vaporizer **102**, e.g., 275° C. These ranges have been found to provide effective vaporization of chemicals such as formic acid and silanes. The cartridge heaters **218**, **234** can be, for example, 250 W cartridge heaters.

(26) In some implementations, as shown in FIG. 3A, the heater slots **232** and holes for the screws **338** (labeled as screw holes **510** in FIGS. 5A-5B) include embossments projecting out from at least some of the plates **304**. The embossments can cause increased mixing of fluid and/or increase the heat transfer surface, both of which can advantageously promote increased vaporization. In some implementations, embossments are provided on at least some of the plates **304** independent of the heater slots **232** and/or screw holes, e.g., one or more embossments (when included) can be separate from the heater slots **232** and/or screw holes.

(27) In some implementations, the heaters, such as the cartridge heaters **218**, **234**, have, as an outer portion (e.g., a sheath), an iron-nickel-chromium alloy, which has been found to provide effective resistance to acidic or corrosive fluids. However, other materials for the heaters are also within the scope of this disclosure, based on material compatibility with the temperature(s) and liquid(s) with which the vaporizer **102** will be used. Further, the heaters are not limited to cartridge heaters, and any one or more suitable types of heater can be used.

(28) In some implementations, further heater(s) are included in the enclosure **206**. For example, multiple outer heaters **344**, shown in FIGS. 3C-3D, can extend longitudinally through the enclosure **206** to surround the baffle assembly **204**. In some implementations, as shown in FIG. 3B, the heaters **344** are inserted from the bottom side of the enclosure **206**. Heater retaining screws **342** can extend longitudinally adjacent to the heaters **344** to hold the heaters **344** in position.

(29) The cover **202** includes an inlet port **208** for liquid flow into the vaporizer **102** (e.g., to the baffle assembly **204** to enter the tortuous path), and an outlet port **210** for vapor flow out of the vaporizer **102**. As shown in FIG. 1, the inlet port **208** can be fluidically coupled to the vessel **104** via a fluid line (e.g., through liquid line **134** and a valve **116**) to receive the liquid. The outlet port **210** can be fluidically coupled to the chamber **112** via a vapor line **126** to provide the vapor to the chamber **112**. The inlet port **208** and the outlet port **210** can include any fitting suitable for liquid

and/or gas transfer, e.g., metal gasket face seal fittings.

(30) It will be understood that the combination, division, and configuration of elements of the vaporizer **102** shown in FIG. 2 (e.g., the elements **202**, **204**, **206**) are examples, and that other implementations are within the scope of this disclosure. For example, in some implementations, at least a portion of the cover **202** is integral with the baffle assembly **204** and/or the enclosure, and/or at least a portion of the baffle assembly **204** is integral with the cover **202** and/or the enclosure **206**. For example, the plates of the baffle assembly **204** (discussed in further detail below) can be formed as integral protrusions from the enclosure **206**. As another example, the baffle assembly **204** can include outer walls that define laterally-outer surfaces of the tortuous path (discussed below), e.g., rather than the outer surfaces being defined by inner walls of the enclosure **206** as in the example of FIG. 2.

(31) As another example, other locations of the inlet port **208** and outlet port **210** are within the scope of this disclosure. For example, the inlet port **208** and the outlet port **210** can be on opposite longitudinal ends of the vaporizer **102** (as opposed to both being at a top side of the vaporizer **102** as in the example of FIG. 2), or at least one of the inlet port **208** or the outlet port **210** can be on a lateral side of the vaporizer **102**. As another example, although the vaporizer **102** of FIG. 2 has a generally tall, narrow, and cylindrical shape, other shapes and proportions of the vaporizer **102** are also within the scope of this disclosure.

(32) As another example, numbers, types, and locations of mechanical coupling elements (e.g., the screws **220**) and the heater(s) (e.g., cartridge heaters **218**, **234**, **244**) can be modified without departing from the scope of this disclosure. In some implementations, at least one heater is configured to heat specifically one or more uppermost plates (closest to the atomizer nozzle), and to not heat one or more plates further from the atomizer nozzle, to ensure that the uppermost plate(s) reach a desired operating temperature in view of the liquid provided thereon.

(33) FIG. 3A is a perspective view of the baffle assembly **204**. FIG. 3B is a cross-sectional view of the vaporizer **102**. These drawings illustrate how the baffle assembly **204** forms a tortuous path for fluid flow to promote highly efficient liquid vaporization.

(34) As shown in FIG. 3B, liquid from the inlet port **208** (e.g., liquid from the vessel **104**) is injected, as a liquid spray **354**, into an interior of the vaporizer **102** by an atomizer including, or consisting of, an atomizer nozzle **352**. The liquid spray **354** includes many small droplets of the liquid, which are projected (in a pressurized injection based on pressure from the pressurizer **106**) onto an opposing surface of a plate **304**. The atomizer promotes vaporization of the liquid in at least two ways. First, the small droplets of the liquid spray **354** have a high surface area, which can make them more prone to vaporization, e.g., compared to a continuous portion of liquid. In some implementations, the atomizer is configured to output the liquid spray **354** with high number of smaller droplets. For example, the atomizer nozzle **352** can be a full-cone nozzle. Second, the liquid spray **354** can be projected such that it contacts a surface opposing the atomizer nozzle **352**—in this example, a plate **304**—over a relatively wide surface area. For example, the plate **304** (or other surface) opposing the atomizer nozzle **352** can be arranged at a distance from the atomizer nozzle **352** (a “spray spacing” as shown in FIG. 3B) that is equal to or close to a focal length of the spread of the liquid spray **354**, e.g., within 10% or within 20% of the focal length. In some implementations, effective spread and vaporization of the injected liquid have been found to be achieved when the spray angle of the liquid spray **354** is in a range from 50° to 80° (e.g., in a range from 60° to 70°), and/or when the spray spacing is in a range from 2 mm to 4 mm (e.g., in a range from 2.5 mm to 3.5 mm).

(35) In some implementations, as shown in FIGS. 3A-3B, an uppermost plate **304** in a stack of plates **304** that receives the liquid spray **354** (in hemisphere **302-1**) is recessed compared to an uppermost plate **304** in another stack of plates **304** (e.g., in hemisphere **302-2**). This positioning provides the spray spacing that permits distribution of the liquid spray **354** over a wide area. For example, the stack of plates in hemisphere **302-1** can have fewer plates than the stack of plates in

hemisphere **302-2**. The uppermost plate **304** can have a surface (e.g., its top surface) exposed to the atomizer nozzle **352**. The use of the atomizer may reduce a concentrated cooling effect caused by liquid hitting the uppermost plate **304** or set of uppermost plates **304**.

(36) The uppermost plate **304** is a plate toward which the liquid spray **354** is directed and can define a first end of the tortuous path **108**. A second end of the tortuous path **108** can be defined, for example, at a last plate **304** along the tortuous path **108** (in the example of FIGS. 3A-3B, an uppermost plate in the stack of plates **304** in hemisphere **302-2**, closest to the outlet port **210**).

(37) In some implementations, effective dispersal of the liquid as small droplets is further aided by the pressure applied to the liquid by the pressurizer **106** that is included in some implementations.

(38) As shown in FIGS. 3A-3B, the baffle assembly **204** includes a series of stacked plates **304** (alternatively referred to as baffles or fins) of two types, plates **304-1** and **304-2**. As described below, the two types of plates **304** in combination (e.g., alternating) form a tortuous path based on different positions of holes on the two types of plates **304**. Two stacks of plates **304**, arrayed parallel to one another and spaced apart by a divider **320** (e.g., a wall or spacer) in which the central heater slot **222** is formed, are arranged in respective hemispheres (alternatively referred to as sections or zones) **302-1**, **302-2** of the baffle assembly **204**. In some implementations, the hemispheres **302-1**, **302-2** are separate components that are attached to one another.

(39) However, the scope of this disclosure is not limited to this arrangement of the plates **304**. For example, in some implementations, only a single stack of plates **304** is present, or three or more stacks of plates are present. For example, in the case of a single stack of plates **304**, fluid flow can be in a top-to-bottom direction with the inlet on the top and the outlet at the bottom, as opposed to the top-to-bottom-to-top flow discussed with respect to FIGS. 3A-3B. As another example, although the vaporizer **102** of FIGS. 3A-3B includes separate plates **304** in the different hemispheres **302-1**, **302-2**, the plates **304** can alternatively be configured to be arranged in both hemispheres **302-1**, **302-2**. Moreover, the baffle assembly **204** need not be divided into hemispheres or other analogous sections. Further, in some implementations, the baffle assembly **204** is divided into more than two sections. For example, the baffle assembly **204** can be divided into four quadrants with dividers between any or all adjacent pairs of the quadrants.

(40) In some implementations, stacks of plates **304** arranged in different hemispheres have different numbers of plates, have different dimensions from one another (e.g., different vertical/longitudinal dimensions, such as height), and/or are vertically offset with respect to one another. For example, different numbers of plates and/or different heights of stacks may facilitate a target spray spacing.

(41) The plates **304** are stacked in the longitudinal direction, and the plate types alternate in the stack, such that the plates **304** include a plate **304-1**, a plate **304-2**, a plate **304-1**, a plate **304-2**, etc., in a direction of the stack. The stack direction, or longitudinal direction, extends along the z-direction in the coordinate axes shown in FIGS. 3A-3D. The z-direction is perpendicular to two lateral directions x and y, which are also perpendicular to one another. Within each stack, the plates **304** extend parallel to one another, having broadest surfaces in the x-y plane. Further, the plates **304** in the two stacks also extend parallel to one another. Sets of screws **338** (or rods or bolts) extend longitudinally through each hemisphere **302-1**, **302-2** (e.g., through screw holes **510** labeled in FIGS. 5A-5B), and nuts **340** corresponding to each plate **304** mate with the screws **338** to hold the plates **304** of each hemisphere **301-1**, **302-2** together. In some implementations, the use of the discrete plates **304** (e.g., joined by the screws **338**) is advantageous for manufacture, as it permits the separate and relatively simple manufacture of each plate **304** separately before the plates **304** are joined together to form the baffle assembly **204**. Further, plates **304** can be replaced (e.g., in the case of damage) relatively easily. However, the plates **304** need not be separate components from one another; in some implementations, at least some of the plates **304** are integral with one another.

(42) In some implementations, at least some of the plates **304** (e.g., adjacent plates within a stack) are inclined relative to one another and/or with respect to the stack direction (longitudinal direction). For example, a first plate may be horizontal while a second plate is tilted, or both plates

may be tilted with respect to the stack direction (e.g., where non-tilted plates are oriented orthogonal to the stack direction). In some implementations, inclined plates promote liquid flow (e.g., downward liquid flow) and/or vapor flow (e.g., upward vapor flow), thereby providing improved fluid flow, vaporization efficiency, and/or overall vapor delivery.

(43) The plates **304** are configured to form and define the tortuous path **108** for flow of the liquid and vapor. The injected liquid (e.g., of the liquid spray **354**) vaporizes and expands, resulting in an increase in pressure that is converted into a flow velocity of the liquid and vapor through the tortuous path **108**. In some implementations, fluid flow through the tortuous path **108** is primarily or entirely caused by vapor expansion, except for any small contribution provided by the initial velocity of the injected liquid. As the liquid and vapor are driven through, or pass through, the tortuous path **108**, the fluid absorbs energy (heat) from the heated plates **304**, such that the liquid is caused to vaporize and the vapor remains in the vapor state (e.g., with little or no condensation). In some implementations, as shown in FIG. 3B, the tortuous path **108** is further defined by inner sidewalls **356** of the enclosure **206** in which the baffle assembly **204** is arranged.

(44) The plates **304** have respective openings **306** (or holes, or cutouts) with positions that increase the length of the tortuous path **108**. For example, a pair of adjacent plates **304** can have respective openings **306** that have different positions relative to an overall flow path (e.g., top-to-bottom or bottom-to-top) of the fluid, such that the fluid is forced through multiple twists, bends, and/or doubling-back paths to travel through the openings **306**. For example, in some implementations, a pair of adjacent plates **304** have respective openings **306** that are at different distances from a longitudinal axis of the vaporizer **102**, such as a central longitudinal axis of the vaporizer **102** (e.g., an axis extending through the central heater slot **222**). In some implementations, a pair of adjacent plates **304**, spaced apart in the z direction, have respective openings **306** that have different x/y coordinates. In some implementations, the openings **306** in a pair of adjacent plates **304** are at entirely different lateral positions. In some implementations, a pair of adjacent plates **304** have at least one opening **306** at a common lateral position, and at least one of the adjacent plates **304** has at least one opening **306** at a lateral position different from lateral positions of any of the openings **306** of the other of the adjacent plates **304**.

(45) In some implementations, as shown for the vaporizer **102** in FIGS. 3A-3B, the openings **306** alternate between inner positions (closer to the central axis of the vaporizer **102**, or radially inward) in plates **304-1**, and outer positions (farther from the central axis, or closer to the lateral outer wall of the vaporizer **102**, or radially outward) in plates **304-2**. Each plate **304-1** has two inner openings **306** (visible in FIG. 3A), and each plate **304-2** has three outer openings (of which one is visible in FIGS. 3A-3B and all are shown diagrammatically in FIG. 3C). The plates **304-1**, **304-2** are shown in a top view in FIGS. 5A and 5B, respectively. The longitudinal axis labeled in FIGS. 5A and 5B corresponds to the central longitudinal axis along which the central heater slot **222** extends. In some implementations, as shown in FIGS. 5A-5B, the plates **304-1**, **304-2** of the two types have the same outer dimensions and the same locations of heater slots **242**.

(46) In the illustrated examples, the baffles or plates are shown as having substantially semicircular shapes. However, other shapes are also within the scope of this disclosure.

(47) The plates **304-1**, **304-2** alternate with one another in each hemisphere **302-1**, **302-2**. As such, the tortuous path **108** repeatedly switches from the radially-outward direction ($-x$ direction) to the radially-inward direction ($+x$ direction) in a sequence of connected chambers defined between adjacent pairs of plates **304**. This configuration contributes to a long tortuous path **108** that is defined in a volume with high surface area. For example, liquid and vapor in the vaporizer **102** cannot simply travel in the vertical $-z$ direction and then in the $+z$ direction to go from the inlet port **208** to the outlet port **210** but, rather, are forced to travel with a long lateral path length in x-y planes, increasing the overall path length and the surface area of the tortuous path **108**.

(48) In addition to or instead of these switches in lateral direction, in some implementations, the tortuous path **108** includes at least one switch in a vertical or longitudinal direction. As shown in

FIG. 3B, the tortuous path **108** first has an overall downward direction ($-z$ direction) in hemisphere **302-1**. The tortuous path **108** then continuous into hemisphere **302-2** (e.g., through an opening **360** as shown in FIG. 3D), where the tortuous path **108** has an overall upward direction ($+z$ direction) to reach the outlet port **210**.

(49) The tortuous path **108** promotes a high vaporization efficiency such that most, all, or substantially all of the injected liquid is converted into vapor, even for relatively high in-flow rates of the liquid. For example, the tortuous path **108** can provide a long fluid flow path relative to the overall dimensions of the vaporizer **102**, increasing vaporization efficiency as the liquid has more time/distance to be vaporized and permitting compact design of the vaporizer **102**.

(50) Further, the tortuous path **108** has a high surface area, corresponding to a thinner layer of the injected liquid distributed across surfaces of the plates **304** and enclosure **206** that define the tortuous path **108**. The thinner layer corresponds to increased vaporization of the liquid.

(51) For example, in some implementations, the baffle assembly **204** is configured such that the tortuous path switches from a first direction to a second direction opposite the first direction (e.g., from radially outward in order to pass through an opening **306** in a plate **304-2**, to radially inward in order to pass through an opening **306** in a plate **304-1**) at least ten times, at least twenty times, at least thirty times, or at least forty times, as shown in FIG. 3B. In some implementations, the baffle assembly **204** is configured such that an overall direction of the tortuous path, neglecting changes in direction associated with individual plates **304**, switches direction at least once, e.g., from the $-z$ direction in hemisphere **302-1** to the $+z$ direction in hemisphere **302-2** as shown in FIG. 3B. In some implementations, the baffle assembly **204** is configured such that, when the tortuous path **108** is defined in a volume having an overall dimension D along a direction, the tortuous path **108** has a length at least $3D$, at least $4D$, or at least $5D$ along the same direction. For example, the baffle assembly **204** can have an inner diameter (between opposite sidewalls **356** of the enclosure **206**) of about 100 mm (which determines a volume for the tortuous path having an overall dimension of 100 mm along the x -axis), and the tortuous path **108** can have a total path length along the radial direction (e.g., $-x$ and $+x$ path segments, or rightward and leftward path segments in FIG. 3B, or radially-inward and radially-outward segments) of at least 500 mm. In some implementations, a distance between adjacent plates **304** in the stack direction (e.g., the z direction of FIGS. 3A-3B) is less than 20 mm or less than 10 mm, e.g., in a range from 3 mm to 7 mm. The spacing between adjacent plates corresponds to a dimension of an intervening space between the vertically-stacked plates **304**. These distances have been found to provide an effective balance between providing a high surface area and long tortuous path **108** (corresponding to a shorter distance), and maintaining a flow resistance that is not excessive (corresponding to a longer distance). In some implementations, a thickness of the plates **304** (e.g., in the stack or z direction) is less than 3 mm, less than 2 mm, or less than 1.5 mm. In some implementations, a lateral distance (e.g., closest distance projected in an x - y plane) between laterally-closest openings **306** in a pair of adjacent plates **304** (e.g., a lateral distance between opening **502** and opening **504** in FIGS. 5A-5B) is at least 10 mm, at least 15 mm, or at least 20 mm. In some implementations, one or more of these characteristics have been found to provide efficient vaporization, e.g., with little or no fluid being un-vaporized when output at the outlet port **210**, even for relatively high liquid throughput in the absence of a carrier gas. This characteristic can be desirable, because liquid provided into the chamber **112** generally does not perform a useful function and represents a decrease in the amount of vapor provided into the chamber **112** for a given amount of liquid provided into the vaporizer **102**.

(52) It will be understood that modifications to the plates **304-1**, **304-2**, the shapes and positions of the openings **306**, and the configuration of the baffle assembly **204** to form a long tortuous path **108**, are within the scope of this disclosure. For example, the plates **304** need not be provided in two types that alternate; rather, in some implementations, different types of plates **304** are arranged in a non-alternating manner, and/or three or more types of plates can be present (e.g., plates **304** of

three types arranged in a pattern of type 1/type 2/type 3/type 1/type 2/type 3, etc.). As another example, although the openings **306** are shown to generally be arranged radially inward and radially outward in the two types of plates **304-1**, **304-2**, the openings **306** can instead or additionally be otherwise arranged to provide a long path length; for example, the openings **306** can be arranged at opposite edges of the plates **304** along the y-axis, without necessarily having different radial positions (though the radial positions can additionally or instead differ, as shown in FIGS. **3A-3B**).

(53) The illustrated configuration of the plates **304** is an example of a configuration of the tortuous path **108** based on sets of openings. For two plates **304-1**, **304-2** that are successive along the tortuous path **108**, each plate **304-1**, **304-2** includes a set of one or more openings **306**. The one or more openings **306** of each set (e.g., of each plate **304-1**, **304-2**) are arranged at a common level along the longitudinal direction (z-axis). For example, the one or more openings **306** can have the same z-coordinate and different x and y coordinates. Further, the one or more openings **306** in a first set (e.g., of plate **304-1**) are arranged at a different level, along the longitudinal axis, than the one or more openings **306** in a second set (e.g., of plate **304-2**). For example, the sets can be arranged at different z-coordinates. Further, the one or more openings **306** of the first set (e.g., of plate **304-1**) are at different lateral positions (e.g., entirely different lateral positions) from the one or more openings of the second set (e.g., of plate **304-2**). It will be apparent that this configuration of openings to define the tortuous path is not reliant on the presence of “plates” but, rather, can be provided using a variety of suitable structures, such as additively-manufactured structures, machined structures, and molded structures.

(54) FIG. **3C** is a diagrammatic top view of the cover **202**, baffle assembly **204**, and enclosure **206**, showing a selection of elements (some of which are internal) and their positions in multiple x-y planes. Heaters **344** are arranged circumferentially around plates **304**, extending longitudinally through the enclosure **206**. Openings **306** in both types of plates **304-1**, **304-2** are illustrated. FIG. **3C** shows an example in which the plates **304-2** have three outer openings **306**, as opposed to four outer openings **306** in the example of FIG. **5B**.

(55) FIG. **3D** is a sectional view illustrating a quarter of the baffle assembly **204**, or half of hemisphere **302-1**. The divider **320** includes an opening **360** through which the tortuous path **108** extends from hemisphere **302-1** into hemisphere **302-2**. The divider **320** can partially define the tortuous path **108**, e.g., define a y-z plane that delimits the tortuous path **108** in the hemisphere **302-1**, as shown in FIG. **3D**. In some implementations, the divider **320** includes two openings **360**, e.g., the opening **360** shown in FIG. **3D** and a second opening at the mirrored position along the y-axis (mirrored across the z-axis).

(56) In some implementations, a configuration of the baffle assembly to have at least two overall chambers (e.g., the hemispheres **302-1**, **302-2** described with respect to FIGS. **3A-3D**) can be advantageous. In the first hemisphere **302-1** into which the liquid is provided, there is a first overall flow direction, e.g., downward. Given a vertical orientation of the baffle assembly **204**, liquid provided into the first hemisphere **302-1** can fall downward with a continuous “waterfall flow” between openings **306** in the plates **304**. In the second hemisphere **302-2**, evaporated vapor rises upward between plates **304**, sustaining the vapor, reducing condensation, and re-evaporating any micro-droplets that may not have fully evaporated in the first hemisphere **302-1**. These effects can be provided, for example, when the vaporizer is configured to provide two overall flow directions that are opposite to one another or substantially opposite to one another (corresponding to both downward and upward overall flow, for example), but can also be provided by other characteristics of the vaporizers described herein.

(57) The heaters illustrated in FIGS. **3A-3D** are configured to maintain the surfaces defining the tortuous path **108** within a target temperature range, e.g., sufficiently high to cause vaporization and prevent condensation. In some implementations, one or more insulating jackets are included to thermally isolate the vaporizer **102** and facilitate its heating. For example, as shown in FIG. **4**, a

cover jacket **402** is configured to fit over the cover **202**, and an enclosure jacket **404** is configured to fit around the enclosure **206**.

(58) FIG. **6** illustrates an example of a vaporizer box **600** in which the vaporizer **102** (shown with jackets **402**, **404**) can be integrated, for example, in the material processing system **100**. The vaporizer box **600** is configured to facilitate liquid delivery to the vaporizer **102** and vapor delivery from the vaporizer **102** to the chamber **112**, while maintaining safety in view of the potentially toxic chemicals handled by the vaporizer **102**. As shown in FIG. **6**, the vaporizer box **600** includes barriers or walls **602**, which in some implementations are at least partially transparent to facilitate monitoring inside the vaporizer box **600**. Liquid line **134** enters the vaporizer box **600** through an opening in a lid **604** and provides liquid to valve **116**, which can be controlled to provide liquid to the vaporizer **102**. Vapor line **126** exits the vaporizer box **600** through the opening or another opening. In some implementations, vapor line **126** is at least partially surrounded by vapor line heater **114** (e.g., a heating jacket), which can be configured to maintain the vapor line **126** at a temperature that prevents condensation of the vapor inside the vapor line **126**, e.g., a temperature in a range from 100° C. to 150° C., such as in a range from 120° C. to 140° C.

(59) A box exhaust outlet **606** is connected to a pump that can continuously remove gas from the vaporizer box **600**, to ensure safety in case of a leak from the vaporizer **102**. An air exhaust intake **608** permits air to flow into the vaporizer box **600** as a result of the exhaust pumping. In some implementations, a leak detector **610** in the vaporizer box **600** is configured to sense a presence of a leak. For example, the leak detector **610** can include a gas sensor configured to detect the level or presence of the vapor produced by the vaporizer **102** (e.g., formic acid vapor, or a vaporous atomic layer deposition or chemical vapor deposition gas), and to output a signal indicative of the level or presence to a controller, e.g., controller **124**. In the event of gas detection, or gas levels above a threshold value, the controller can shut down the system, e.g., turn off the valve **116** to prevent production of more vapor. In some implementations, a lid switch **614** is configured to sense whether the lid is open or closed. For example, the lid switch **614** can provide a signal to the controller or an interlock device, and the controller or interlock device can permit liquid flow to the vaporizer **102** only when the lid **604** is closed, for safety, e.g., by controlling the valve **116**. An electrical panel **612** is configured to provide signal and power inputs/outputs to components in the vaporizer box **600**, e.g., sensor signals, heater control signals, heater power signals, and/or valve control signals.

(60) FIG. **7** illustrates examples of the vessel **104** and the pressurizer **106**. In some implementations, the vessel **104** includes one or more sensors **712**, e.g., a leak sensor and/or a level sensor. The pressurizer **106** includes a regulator **704** (e.g., an adjustable regulator) connected between a gas inject line **706** (fluidically coupling the regulator **704** to the vessel **104**) and a gas supply line **706** (fluidically coupling the regulator **704** to a gas tank). Gas from the gas tank is injected into the vessel **104** at a pressure set by the regulator **704** (e.g., an adjustable pressure controlled by signals provided by the controller **124**), to pressurize liquid **702** in the vessel **104** and push the liquid **702** into the vaporizer **102** (e.g., through the valve **116**). In some implementations, the gas is an inert gas such as nitrogen (N.sub.2) or argon (Ar). In some implementations, the gas inject line **706** is arranged such that the gas is injected into a space above the liquid **702**. For example, a port of the vessel **104** for coupling to the gas inject line **706** can be arranged at a top of the vessel **104** or otherwise above a surface level of the liquid **702**. As such, the gas can provide pressure without itself being provided into the vaporizer **102**. In some implementations, the liquid line **134** is arranged such that the liquid **702** is above the liquid line **134**, so as to be pushed down into the liquid line **134**. For example, a port of the vessel **104** for coupling to the liquid line **134** can be at a bottom of the vessel **104** or otherwise below a lowest level of the liquid **702**.

(61) In some implementations, the “push pressure” of the injected gas is in a range from 35 psi to 90 psi, such as a range from 20 psi or 40 psi to 60 psi. These pressures have been found to provide consistent delivery of the liquid **702** to the vaporizer **102**. The push pressure can be greater than

atmospheric pressure. In some implementations, the pressurizer **106** includes a pressure sensor **710** (e.g., in the vessel **104**, in a fluid line connected to the vessel **104**, or integrated with the regulator **704**) configured to detect the push pressure. In some implementations, the pressure sensor provides an output signal, indicative of the sensed pressure, to a controller (e.g., controller **124**), and the controller can control the regulator **704** to set the push pressure to a target value or to be within a target range. This control can provide precise control of the push pressure, which in some implementations can facilitate precise liquid delivery (corresponding to precise vapor delivery to the chamber **112**).

(62) The valve **116** can be any suitable valve for regulating liquid flow to the vaporizer **102**. In some implementations, the valve **116** is a fast-acting valve, e.g., with a controllable on-time. In some implementations, the valve **116** can be controlled (e.g., by the controller **124**) to pulse with an adjustable on-time for each pulse, to provide pulsed volume injection control. Additional details on control of the valve **116** are provided below with respect to FIG. **8**.

(63) For purposes of this disclosure, it has been determined that the pressurized provision of liquid can provide high and precisely-controlled levels of liquid injection into the vaporizer **102**. This liquid injection can correspond to high rates of vapor delivery to the chamber **112**. Further, in some implementations, the pressurizer **106** acts in conjunction with the atomizer **110** to generate the liquid spray **354** that facilitates high vaporization efficiency.

(64) In some implementations, the vaporizer **102** can be used and/or is configured to deliver vapor with little or no carrier gas. For example, gas output from the vaporizer **102** can be at least 75%, at least 90%, at least 90%, at least 95%, or at least 99% composed of vapor of the liquid provided into the vaporizer **102**. In some implementations, the vapor output from the vaporizer **102** is substantially entirely composed of the vapor of the liquid. In some implementations, the vaporizer **102** includes only a single fluid inlet (e.g., the inlet port **208** for liquid), and/or is configured or operated such that only the single fluid inlet is used. As such, vapor dilution can be reduced or eliminated, and the vapor can be delivered more effectively, e.g., with a higher partial pressure of the vaporized liquid.

(65) FIG. **8** illustrates an example of a process **800** for vapor delivery. The process **800** can be performed, for example, by the controller **124**. The process **800** includes setting a target vaporizer temperature and/or a target push pressure (**802**). For example, the controller **124** can send signals to one or more of the cartridge heaters **218**, **234**, **344** to set an internal temperature of the vaporizer **102**, e.g., to be in a range from 160° C. to 220° C. The controller **124** can send the signals based on temperature signals from one or more temperature sensors in the vaporizer **102**, e.g., a thermocouple disposed in the central heater slot **222**. The controller **124** can send a signal to the regulator **704** to set the push pressure, e.g., to be in a range from 35 psi to 90 psi, in some implementations based on a pressure sensed by the pressure sensor **710**.

(66) The process **800** includes, when the temperature and/or pressure are within target range(s), pulsing liquid into the vaporizer for an on-time (**804**). For example, the controller **124** can send a signal to the valve **116** to cause the valve **116** to pulse open. In some implementations, the on-time is in a range from 20 ms to 250 ms. In some implementations, the on-time is in a range from 40 ms to 250 ms, in a range from 40 ms to 150 ms, or in a range from 40 ms to 100 ms, time ranges which have been found to provide target delivery rates and target pressure increases in the chamber **112**. For example, pulses from 40 ms to 250 ms, in conjunction with the vaporizers **102** described herein, have been found to consume liquid at a rate between 1.6 g/pulse and 2.3 g/pulse, a rate compatible with pressure requirements for the chamber **112**.

(67) After the pulsed delivery, the valve is maintained off for an off-time (**806**). For example, the controller **124** can pause after causing the prior pulse (**804**), before sending the next pulse, for the off-time. In some implementations, it has been found that the off-time is an important configurable parameters. The off-time allows liquid pressure to build back up behind the valve **116**. However, longer off-times may result in slower vapor delivery rates, because pulses of liquid are delivered to

the vaporizer **102** at a slower rate. In some implementations, the off-time is at least three times the on-time, at least five times the on-time, at least ten times the on-time, at least twenty times the on-time, or at least forty times the on-time. In some implementations, the off-time is in a range from 0.5 seconds to 4.0 seconds. These off-times have been found to result in target push-pressure recovery and target vapor delivery rates.

(68) Operations **804** and **806** can be repeated, for example, a predetermined number of times, until the chamber **112** reaches a target pressure, or until a process in the chamber **112** is completed. During repetition of operations **804** and **806**, in some implementations, the temperature and/or pressure are monitored, and the controller **124** may provide control signals responsive to measured temperature and/or pressure to cause the temperature and/or pressure to be within their target range(s).

(69) FIG. **9** illustrates an example of a process **900** that can be performed using the vaporizers described herein, e.g., using the materials processing system **100**. The process **900** is described using the example of a solder reflow process using formic acid. However, it will be understood that the systems described herein are equally applicable to other chemical vaporizers and/or processes. For example, in some implementations, the vapor generated by the vaporizer **102** is an etchant, for use in etch processes. In some implementations, the vapor generated by the vaporizer **102** is a chemical vapor deposition or atomic layer deposition precursor, for use in deposition processes. Further, the characteristics of the materials processing system **100** described in conjunction with process **900** are not dependent on the process **900** but, rather, can be included in the materials processing system **100** independent of process(es) performed using that system.

(70) As shown in FIG. **9**, the process **900** includes injecting a pressurized liquid into a vaporizer, such as vaporizer **102** (**902**). For example, the liquid can be formic acid. The liquid can be pressurized using a pressurizer **106** as discussed in reference to FIG. **7**. Referring to FIG. **1**, the liquid can be injected from the vessel into the vaporizer **102** through liquid line **134** and valve **116**. In some implementations, the liquid is injected into the vaporizer **102** through an such as atomizer **110**, e.g., to form a liquid spray in the vaporizer **102** to promote vaporization. In some implementations, the vaporizer includes a tortuous path, such as tortuous path **108**, that promotes vaporization with a high vaporization efficiency. In some implementations, process **900** includes process **800**, e.g., temperature and/or pressure control in conjunction with pulsed liquid injection.

(71) The process **900** includes providing a vapor of the liquid (generated using the vaporizer) into a chamber (**904**). For example, as shown in FIGS. **1** and **2**, the vapor can be output through outlet port **210** and provided into chamber **112** through the vapor line **126**. As discussed in reference to FIG. **6**, in some implementations the vapor line **126** is at least partially surrounded by a vapor line heater **114** that is operated or controlled to heat the vapor line heater.

(72) Referring to FIG. **1**, in some implementations, the chamber **112** includes a pressure sensor **122** configured to sense an internal pressure of the chamber **112**. In some implementations, signals from the pressure sensor **122** are provided to a controller (e.g., controller **124**), and the controller, based on the chamber pressure, adjusts one or more process parameters, such as push pressure, pulse on-time, and/or pulse off-time, in order to achieve a target pressure in the chamber **112**. Instead or additionally, the pressure sensor **122** can be connected to a switch **140** that enables or disables one or more components of the material processing system **100** based on the pressure. For example, the switch **140** can enable the valve **116** (and/or another component, e.g., an outlet valve of a tank holding the gas used for pressurization) only if the chamber pressure is below a threshold value, e.g., 0.5 atm. The switch **140** can improve safety by preventing outgassing from the chamber **112**.

(73) The chamber **112** can be connected to a pump **118** via a foreline **142**. The pump **118** can be configured to remove vapor from the chamber **112**. In some implementations, one or more valves can enable/disable pumping of the chamber **112** by the pump **118**. In some implementations, a foreline heater **120** at least partially surrounds the foreline **142**. The foreline heater **120** (e.g., a heating jacket), which can be configured to maintain the foreline **142** at a temperature that prevents

condensation of the vapor inside the foreline **142**, e.g., a temperature in a range from 100° C. to 150° C., such as in a range from 120° C. to 140° C.

(74) The chamber **112** includes a sample chuck **132** for holding a sample **130** on which processing is to be performed using vapor from the vaporizer **102**. In some implementations, the chamber **112** includes one or more chamber heaters **144** that can be operated (e.g., by the controller **124**) to maintain the internal temperature of the chamber within a target range. In some implementations, the chamber **112** includes a chamber temperature sensor **136** for outputting signals indicative of the internal temperature, e.g., to the controller **124** for use in controlling the chamber heaters **144**.

(75) The chamber **112** can additionally include further sample-handling and/or other components such as load-lock(s), transfer arm(s), a vent inlet, and/or the like. Also, it will be understood that the illustration of the material processing system **100** in FIG. **1** does not exclude the presence of additional or alternative components, such as additional valves, sensors, fluid-handling components, sample-handling components, gas sources, and/or the like. In some implementations, the material processing system **100** includes multiple vaporizers **102** connected to the same and/or different sources of liquid and connected to the chamber **112** and/or one or more other chambers.

(76) Continuing in reference to FIG. **9**, the process **900** includes performing material processing in the chamber using the vapor (**906**). For example, in some implementations, the vapor is formic acid vapor, and the sample **130** includes components to be bonded together using solder. For example, the sample can include a glass panel with tin contacts, the tin contacts to be bonded to chips. The controller **124** can control the material processing system **100** to maintain the chamber temperature in a range suitable for reflowing (e.g., at about 225° C., with the understanding that various different chamber temperatures can be used for various processes), and can control the valve **116** to deliver formic acid into the chamber **112**. The formic acid vapor removes the native oxide from the tin surface. In some implementations, during solder reflow, the formic acid pressure in the chamber **112** is maintained in a range from 125 Torr to 175 Torr, with the understanding that various different chamber pressures can be used for various processes.

(77) Materials of the material processing system **100** can be compatible with the liquid and vapor provided into and out of the vaporizer **102**. In some implementations, the vaporizer **102** (e.g., plates **304**, cover **202** and/or enclosure **206**) are composed of aluminum and/or steel. In some implementations, the atomizer nozzle **352** is composed of steel. In some implementations, fluid lines of the material processing system **100** (e.g., lines **134**, **126**, and/or **120**) are composed of steel. In some implementations, a diaphragm of the valve **116** is composed of a cobalt-based superalloy compatible with corrosive environments. In some implementations, the vessel **104** is composed of steel. Steel used in the material processing system **100** can be resistant to corrosion, e.g., corrosion by formic acid. For example, in some implementations, the steel includes 316L stainless steel.

(78) As such, based on the foregoing systems, apparatuses, and processes, vapor may be generated and provided with a high flow-rate and with a high vaporization efficiency. For example, the use of pressurizers, atomizers, and/or tortuous paths can promote vaporization. In some implementations, the described vaporizers include long tortuous paths in an overall compact vaporizer to improve system form-factor. Further, in some implementations, the use of plates to form the tortuous path provides improved ease of manufacturing and/or repair. The delivered vapor can be provided with a high flow-rate for a corresponding high process throughput. By contrast, some alternative vaporizers are non-compact, which may have negative effects both in terms of general spatial efficiency (e.g., given limited space for processing tools) and for vapor provision in particular, because a close vaporizer-chamber distance may be desirable in order to reduce cooling, condensation, and/or other negative results of longer piping.

(79) The term “computing system” as used in this disclosure may encompass all apparatus, devices, and machines for processing data, including by way of example a programmable processor, a computer, or multiple processors or computers. A processing system, such as the controller **124**, can include, in addition to hardware, code that creates an execution environment for the computer

program in question, e.g., code that constitutes processor firmware, a protocol stack, a database management system, an operating system, or a combination of one or more of them.

(80) A computer program (also known as a program, software, software application, script, executable logic, or code) can be written in any form of programming language, including compiled or interpreted languages, or declarative or procedural languages, and it can be deployed in any form, including as a standalone program or as a module, component, subroutine, or other unit suitable for use in a computing environment. A computer program does not necessarily correspond to a file in a file system. A program can be stored in a portion of a file that holds other programs or data (e.g., one or more scripts stored in a markup language document), in a single file dedicated to the program in question, or in multiple coordinated files (e.g., files that store one or more modules, sub programs, or portions of code). A computer program can be deployed to be executed on one computer or on multiple computers that are located at one site or distributed across multiple sites and interconnected by a communication network.

(81) Computer readable media suitable for storing computer program instructions and data include all forms of non-volatile or volatile memory, media and memory devices, including by way of example semiconductor memory devices, e.g., EPROM, EEPROM, and flash memory devices; magnetic disks, e.g., internal hard disks or removable disks or magnetic tapes; magneto optical disks; and CD-ROM and DVD-ROM disks. The processor and the memory can be supplemented by, or incorporated in, special purpose logic circuitry. Sometimes a server is a general-purpose computer, and sometimes it is a custom-tailored special purpose electronic device, and sometimes it is a combination of these things.

(82) Implementations can include a back end component, e.g., a data server, or a middleware component, e.g., an application server, or a front end component, e.g., a client computer having a graphical user interface or a Web browser through which a user can interact with an implementation of the subject matter described is this specification, or any combination of one or more such back end, middleware, or front end components. The components of the system can be interconnected by any form or medium of digital data communication, e.g., a communication network. Examples of communication networks include a local area network (“LAN”) and a wide area network (“WAN”), e.g., the Internet.

(83) The features described for the controller **124** can be implemented in digital electronic circuitry, or in computer hardware, firmware, software, or in combinations of them. The apparatus can be implemented in a computer program product tangibly embodied in an information carrier, e.g., in a machine-readable storage device, for execution by a programmable processor; and method steps can be performed by a programmable processor executing a program of instructions to perform functions of the described implementations by operating on input data and generating output. The described features can be implemented advantageously in one or more computer programs that are executable on a programmable system including at least one programmable processor coupled to receive data and instructions from, and to transmit data and instructions to, a data storage system, at least one input device, and at least one output device. A computer program is a set of instructions that can be used, directly or indirectly, in a computer to perform a certain activity or bring about a certain result. A computer program can be written in any form of programming language, including compiled or interpreted languages, and it can be deployed in any form, including as a stand-alone program or as a module, component, subroutine, or other unit suitable for use in a computing environment.

(84) A number of implementations have been described. Nevertheless, it will be understood that various modifications may be made.

(85) While this disclosure contains many specific implementation details, these should not be construed as limitations on the scope of any invention or of what may be claimed, but rather as descriptions of features that may be specific to particular implementations of particular inventions. Certain features that are described in this disclosure in the context of separate implementations can

also be implemented in combination in a single implementation. Conversely, various features that are described in the context of a single implementation can also be implemented in multiple implementations separately or in any suitable subcombination. Moreover, although features may be described above as acting in certain combinations and even initially claimed as such, one or more features from a claimed combination can in some cases be excised from the combination, and the claimed combination may be directed to a subcombination or variation of a subcombination. (86) Similarly, while operations are depicted in the drawings in a particular order, this should not be understood as requiring that such operations be performed in the particular order shown or in sequential order, or that all illustrated operations be performed, to achieve desirable results.

Claims

1. A vaporizer apparatus comprising: a liquid inlet; a plurality of plates in a stacked arrangement, the plurality of plates comprising openings and surfaces defining a tortuous path for a liquid through the plurality of plates, wherein the plurality of plates is configured to direct a flow of the liquid through the tortuous path from a first end corresponding to a first plate of the plurality of plates and output a vapor of the liquid at a second end of the tortuous path corresponding to a last plate of the plurality of plates; and at least one heating element configured to heat the plurality of plates to vaporize the liquid along its flow through the tortuous path and generate the vapor, wherein the plurality of plates comprises a first plurality of plates stacked in a first section, and a second plurality of plates stacked in a second section, wherein the first plurality of plates is laterally spaced apart from the second plurality of plates, wherein the first plurality of plates comprises openings and surfaces defining a first portion of the tortuous path, and wherein the second plurality of plates comprises openings and surfaces defining a second portion of the tortuous path.
2. The vaporizer apparatus of claim 1, wherein the plurality of plates are stacked with intervening spaces having a specified dimension.
3. The vaporizer apparatus of claim 1, wherein the plurality of plates comprise two plates in succession along the tortuous path, wherein the two plates are oriented parallel to one another and spaced apart along a stack direction, and wherein the two plates comprise respective openings arranged at different lateral positions.
4. The vaporizer apparatus of claim 3, wherein the respective openings of one of the two plates are arranged more radially inward compared to the respective openings of another of the two plates.
5. The vaporizer apparatus of claim 1, comprising an atomizer configured to atomize liquid from the liquid inlet and inject a spray of the liquid toward the first plate of the plurality of plates.
6. The vaporizer apparatus of claim 1, wherein the first plate is included in the first plurality of plates and is closest to the liquid inlet among plates of the first plurality of plates, and wherein the last plate is included in the second plurality of plates and is closest to the liquid inlet among plates of the second plurality of plates.
7. The vaporizer apparatus of claim 1, comprising a divider separating the first plurality of plates from the second plurality of plates, wherein the divider comprises an opening that defines a portion of the tortuous path.
8. The vaporizer apparatus of claim 1, comprising a divider separating the first plurality of plates from the second plurality of plates, wherein the divider comprises a slot to receive a first heating element of the at least one heating element.
9. The vaporizer apparatus of claim 1, wherein: (i) the first plurality of plates and the second plurality of plates include different numbers of plates, (ii) a total dimension of the first plurality of plates along a stack direction is different from a total dimension of the second plurality of plates along the stack direction, or (i) and (ii).
10. The vaporizer apparatus of claim 1, comprising an enclosure in which the plurality of plates are

arranged, wherein an inner sidewall of the enclosure defines the tortuous path in conjunction with the plurality of plates.

11. The vaporizer apparatus of claim 1, wherein the first portion of the tortuous path extends from the first end of the tortuous path with a first overall flow direction; and wherein the second portion extends from the first portion to the second end of the tortuous path with a second overall flow direction, wherein the second overall flow direction is opposite to the first overall flow direction.

12. The vaporizer apparatus of claim 1, wherein the tortuous path switches from a first direction to a second direction opposite the first direction at least twenty times.

13. The vaporizer apparatus of claim 1, wherein a length, along a lateral direction, between opposite lateral edges of a volume defining the tortuous path is D , and wherein the tortuous path has a length at least $3D$ along the lateral direction.

14. The vaporizer apparatus of claim 1, wherein the vaporizer apparatus is configured to deliver the vapor to a chamber in the absence of a carrier gas.

15. The vaporizer apparatus of claim 1, wherein gas output from the vaporizer apparatus is at least 95% vapor of the liquid.

16. The vaporizer apparatus of claim 1, wherein the liquid inlet is the only fluid inlet of the vaporizer apparatus.

17. The vaporizer apparatus of claim 1, wherein the plurality of plates are separate components from one another.

18. A vaporizer apparatus comprising: a liquid inlet; a plurality of plates in a stacked arrangement, the plurality of plates comprising openings and surfaces defining a tortuous path for a liquid through the plurality of plates, wherein the plurality of plates is configured to direct a flow of the liquid through the tortuous path from a first end corresponding to a first plate of the plurality of plates and output a vapor of the liquid at a second end of the tortuous path corresponding to a last plate of the plurality of plates; and at least one heating element configured to heat the plurality of plates to vaporize the liquid along its flow through the tortuous path and generate the vapor, wherein the plurality of plates comprise a first type of plate and a second type of plate, wherein the first type of plate and the second type of plate have openings in different lateral positions, and wherein the first type of plate and the second type of plate alternate along the tortuous path.

19. The vaporizer apparatus of claim 18, wherein the openings in the first type of plate are arranged more radially inward compared to the openings in the second type of plate.

20. The vaporizer apparatus of claim 19, wherein the at least one heating element comprises a heating element extending through two or more of the plurality of plates.

21. The vaporizer apparatus of claim 18, comprising an atomizer configured to atomize liquid from the liquid inlet and inject a spray of the liquid toward the first plate.

22. The vaporizer apparatus of claim 18, wherein the vaporizer apparatus is configured to deliver the vapor to a chamber in the absence of a carrier gas.

23. A vaporizer apparatus comprising: a liquid inlet; a plurality of plates in a stacked arrangement, the plurality of plates comprising openings and surfaces defining a tortuous path for a liquid through the plurality of plates, wherein the plurality of plates is configured to direct a flow of the liquid through the tortuous path from a first end corresponding to a first plate of the plurality of plates and output a vapor of the liquid at a second end of the tortuous path corresponding to a last plate of the plurality of plates; and at least one heating element configured to heat the plurality of plates to vaporize the liquid along its flow through the tortuous path and generate the vapor, wherein the at least one heating element comprises a cartridge heater extending through two or more of the plurality of plates.

24. The vaporizer apparatus of claim 23, wherein the plurality of plates are arranged in a first stack and a second stack spaced laterally apart from the first stack, and wherein the cartridge heater extends through all plates in the first stack or the second stack.

25. The vaporizer apparatus of claim 23, wherein the plurality of plates comprise two plates in

succession along the tortuous path, wherein the two plates are oriented parallel to one another and spaced apart along a stack direction, and wherein the two plates comprise respective openings arranged at different lateral positions.

26. The vaporizer apparatus of claim 23, wherein the tortuous path comprises: a first portion extending from the first end of the tortuous path with a first overall flow direction; and a second portion extending from the first portion to the second end of the tortuous path with a second overall flow direction, wherein the second overall flow direction is opposite to the first overall flow direction.

27. A vaporizer apparatus comprising: a liquid inlet; a plurality of plates in a stacked arrangement, the plurality of plates comprising openings and surfaces defining a tortuous path for a liquid through the plurality of plates, wherein the plurality of plates is configured to direct a flow of the liquid through the tortuous path from a first end corresponding to a first plate of the plurality of plates and output a vapor of the liquid at a second end of the tortuous path corresponding to a last plate of the plurality of plates; and at least one heating element configured to heat the plurality of plates to vaporize the liquid along its flow through the tortuous path and generate the vapor, wherein the vaporizer apparatus is configured to deliver the vapor to a chamber in the absence of a carrier gas, wherein the plurality of plates comprise two plates in succession along the tortuous path, wherein the two plates are oriented parallel to one another and spaced apart along a stack direction, and wherein the two plates comprise respective openings arranged at different lateral positions.

28. The vaporizer apparatus of claim 27, wherein gas output from the vaporizer apparatus is at least 95% vapor of the liquid.

29. The vaporizer apparatus of claim 27, wherein the liquid inlet is the only fluid inlet of the vaporizer apparatus.
